Estimating Population

- The term population is used to indicate the total number of human beings in a certain area at any particular time.
- The present population is obtained by referring to the statistics of census records prepared by the local body.
- The water supply projects is not designed only for present population. But it is made to accommodate the future population.
- The growth of population may be sharp, slow or even stationary depending upon the factors contributing to the future development of the locality such as coming up of new industries, trade expansion, etc.
- The future period for which various services units of water supply or sanitary engineering are designed is known as the *period of design.*
- The period of design may varies from 20 to 40 years or even upto 50 years.

POPULATION FORECASTING

Design of water supply and sanitation scheme is based on the projected population of a particular city, estimated for the design period.

Any underestimated value will make system inadequate for the purpose intended; similarly overestimated value will make it costly. Changes in the population of the city over the years occur, and the system should be designed taking into account of the population at the end of the design period.

Factors affecting changes in population are:

- increase due to births
- decrease due to deaths
- increase/ decrease due to migration
- increase due to annexation.

POPULATION FORECASTING METHODS

- Arithmetic Increase method
- Geometric Increase Method
- Incremental Increase Method
- Decrease Rate of growth Method
- Simple Graphical Method
- Comparitive Graphical Method
- Zoning method or master plan method
- Ratio and correlation method
- Growth composition analysis method

AIRTHMETIC INCREASE METHOD

- This is the most simple method of population forecast , through it generally gives lower results.
- This method is suitable for large and old city with considerable development.
- If it is used for small, average or comparatively new cities, it will give lower population estimate than actual value.
- In this method the average increase in population per decade is calculated from the past census reports.
- This increase is added to the present population to find out the population of the next decade.
- Thus, it is assumed that the population is increasing at constant rate.

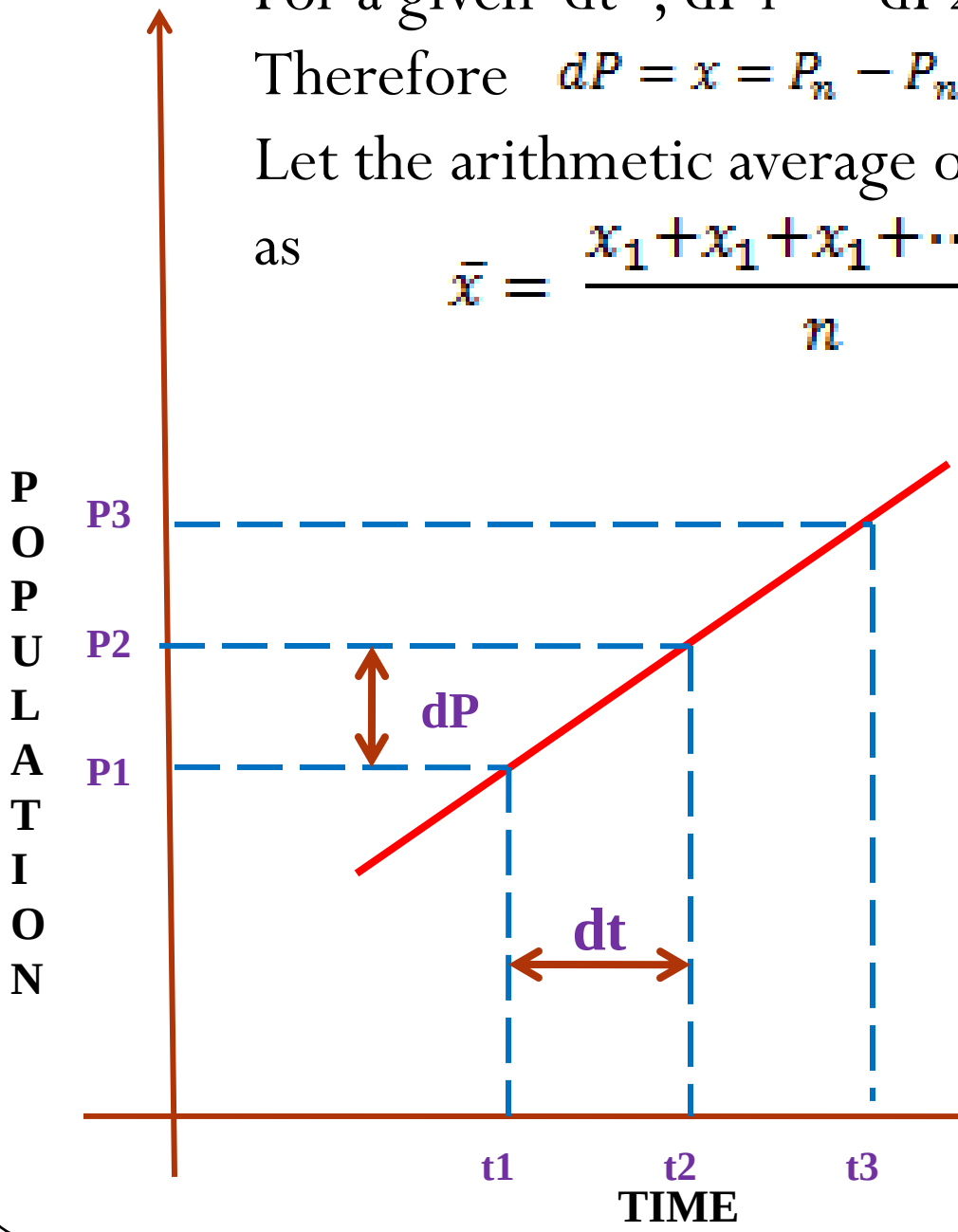
Hence, $dP/dt = K$ i.e., rate of change of population with respect to time is constant.

For a given ‘dt’ , dP1 = dP2 = dP3 = constant

Therefore $dP = x = P_n - P_{n-1}$

Let the arithmetic average of per decade increases is given as

$$\bar{x} = \frac{x_1 + x_1 + x_1 + \cdots x_n}{n}$$



P0 be the latest known last decades population.

Population after 1 decade:

$$P_1 = P_0 + 1\,X\,\bar{x}$$

Population after 2 decade:

$$P_2 = P_0 + 2\,X\,\bar{x}$$

Therefore formulae derived is

$$\mathbf{P_n = P_0 + n\,\bar{x}}$$

Where ‘n’ = no. of decades

Example. The population figure in a growing town are as follows. The predicted population in 2010 by arithmetic increase method

YEAR	1970	1980	1990	2000
POPULATION	40000	46000	53000	58000

Sol:

Year	Population	Per decade increases in population
1970	40000	6000
1980	46000	7000
1990	53000	5000
2000	58000	

$$\bar{x} = \frac{6000 + 7000 + 5000}{3} = 6000$$

$$P_n = P_0 + n\bar{x} \qquad n = \frac{2010 - 2000}{10} = 1$$

$$P_0 = 58000$$

$$P_{2010} : P_1 = P_0 + 1\bar{x} = 58000 + 1(6000) = 64000$$

Example: 1

Predict the population for the year 2021, 2031, and 2041 from the following population data.

Year	1961	1971	1981	1991	2001	2011
Population	8,58,545	10,15,672	12,01,553	16,91,538	20,77,820	25,85,862

Solution

Year	Population	Increment
1961	858545	-
1971	1015672	157127
1981	1201553	185881
1991	1691538	489985
2001	2077820	386282
2011	2585862	508042

Average increment = 345463

Population forecast for year 2021 is, $P_{2021} = 2585862 + 345463 \times 1 = 2931325$

Similarly, $P_{2031} = 2585862 + 345463 \times 2 = 3276788$

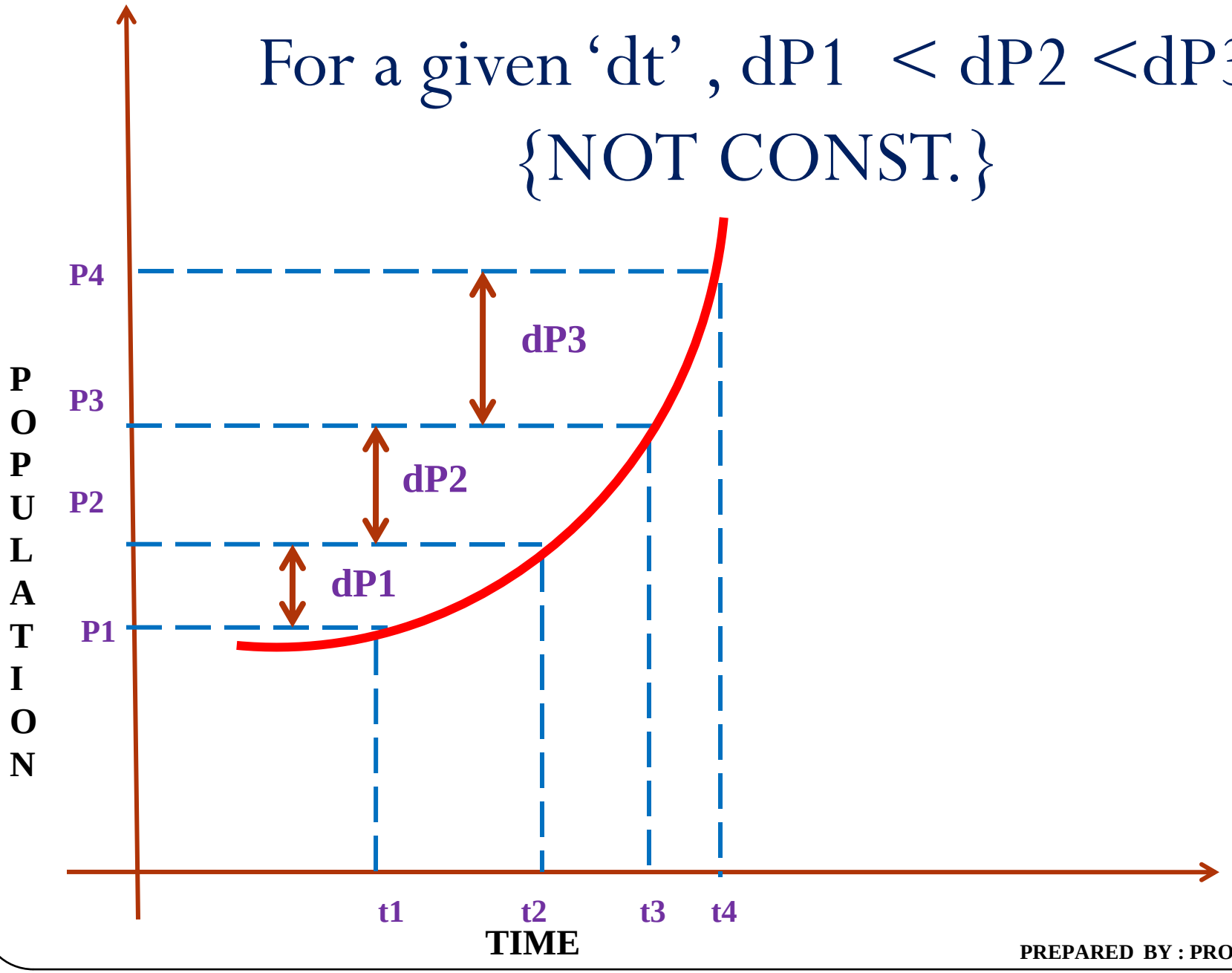
$P_{2041} = 2585862 + 345463 \times 3 = 3622251$

GEOMETRIC INCREASE METHOD

- This method is suitable for fast and rapidly developing cities.
- In this method percentage increase in population from decade to decade is assumed to be remain constant.
- Since this method give higher values and hence should be applied for a new industrial town at the beginning of development for only few decades.
- Thus, it is assumed that the per decade % growth rate is assume to be constant.

$$\frac{dP}{dt} \propto P_t \longrightarrow \frac{dP}{dt} = r.P_r$$

For a given ‘dt’ , dP1 < dP2 <dP3 {NOT CONST.}



$$P_{t_n} = P_0 . e^{r[t_n - t_0]}$$

When geometric percentage increases per decades is given directly in problems

When geometric percentage increases per decades is not given in problems the it is generated by using population statistics of community.

Let $\bar{r}$ be the gemetric Average of per decade % increases

$$\bar{r} = (r_1 * r_2 * r_3 \ldots\ldots\ldots r_n)^{1/n} \qquad r_n = \frac{P_n - P_{n-1}}{P_{n-1}} * 100$$

To find of future population 'r' is compounded to the existing population.

Population after 1 decade is $P_1 = P_0 * [1 + \frac{r}{100}]$

Population after 'n' decade

$$\mathbf{P_n = P_0 * [1 + \frac{r}{100}]^n}$$

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Example. The population figure in a growing town are as follows. The predicted population in 2030 by Geometric increase method

YEAR	1980	1990	2000	2010
POPULATION	40000	50000	80000	140000

Sol:

Year	Population	Per decade % increases in population
1980	40000	[(50-40)/40]×100 = 25
1990	50000	[(80-50)/50]×100 = 60
2000	80000	[(140-80)/80]×100=75
2010	140000	

$$n = \frac{2030 - 2010}{10} = 2\ decades \qquad P_0 = 140000$$

Geometric average of per decade % increase in population $(\bar{r}) = (r_1 \times r_2 \times r_3)^{\frac{1}{3}} = (25 \times 60 \times 75)^{\frac{1}{3}}$

$$= 48.27\%$$

$$P_n = P_{2030} = P_0\left[1 + \frac{\bar{r}}{100}\right]^n = 140000\left[1 + \frac{48.27}{100}\right]^2$$

$$= 307795$$

The population of a town in three consecutive decades are 1 lakh , 1.4 lakh , 1.68 lakh respectively . The population of this town in the fourth consecutive decade according to geometric method would be?

time	Population (lakh)	% Increased
1	1	
2	1.4	$\left(\frac{1.4-1}{1}\right) \times 100 = 40\%$
3	1.68	$\left(\frac{1.68-1.4}{1.4}\right) \times 100 = 20\%$

$P_0 = 1.68$ Lakh , $\bar{r} = (r_1 \times r_2)^{1/2} = (40 \times 20)^{1/2} = 28.28\%$

$$P_n = P_0 * \left[1 + \frac{r}{100}\right]^n$$

$n = 1$, $P_0 = 1.68$

$$\therefore P_4 = 1.68 \left[1 + \frac{28.28}{100}\right]^1 = 2.45 \text{ lakh}$$

INCREMENTAL INCREASE METHOD

- This method is modification of AIM .
- It is suitable for average town under normal condition where the growth rate is to increasing order.

For a given 'dt' , dP1 < dP2 <dP3

OR

For a given 'dt' , dP1 >dP2 > Dp3

- Future population by this method base on average of the increments over increase is +ve or –ve.
- Let $\bar{y}$ be the avg. of increment over increases to find future population by this method is $\bar{y}$ is added to Avg.Increases

Population after 1 decade ,

$$P_1 = P_o + 1 * \bar{x} + \frac{1 * (1+1)}{2} * \bar{y} \longrightarrow P_1 = P_0 + \bar{x} + 1 * \bar{y}$$

Population after 2 decade ,

$$P_2 = P_o + 2 * \bar{x} + \frac{2 * (2+1)}{2} * \bar{y}$$

$$P_2 = P_0 + 2 * \bar{x} + 3 * \bar{y}$$

Population after ‘n’ decade ,

$$P_n = P_o + n * \bar{x} + \frac{n * (n+1)}{2} * (\pm \bar{y})$$

Where,

$\bar{x} = per\ decade\ increase\ in\ population$

$\bar{y} = avg.\ incremental\ increase\ in\ population$

Past population data of a town is given below. Find The population in the year 2030 incremental increase method.

Year	Population
1980	25000
1990	40000
2000	50000
2010	70000

Sol:

Year	Population	Per decade increases in population	Increment increase over in population
1980	25000	15000	– 5000
1990	40000	10000	+10000
2000	50000	20000	
2010	70000		

$$\bar{x} = \frac{15000 + 10000 + 20000}{3} = 15000$$

$$\bar{y} = \frac{-5000 + 10000}{2} = +2500$$

$$P_0 = 70000\,;\; n = \frac{2030 - 2010}{10} = 2\ decades$$

$$P_n = P_0 + n\bar{x} + \frac{n(n+1)}{2}\bar{y}$$

$$P_{2030} = P_2 = 70000 + 2(15000) + \frac{2(2+1)}{2}(+2500)$$

$$= 107500$$

Given the population figure of place is 50000 in 1990 . The average increase in population is 5000 per decade and average incremental increase is 500 per decade. Then expected population in 2020 will be?

$$P_n = P_o + n * \bar{x} + \frac{n * (n+1)}{2} * (\pm\bar{y})$$

$P_{1990} = 50000$

$\bar{x} = 5000$ per decade

$\bar{y} = 500$ per decade

$P_{2020} = ?$

$n = \frac{2020 - 1990}{10}$

$\therefore n = 3$

$P_{2020} = 50000 + 3 \times 5000 + \frac{3(3+1)}{2} \times 500$

$P_{2020} = 68000$

L-U

The population of a town is shown below:

Year	Population	Year	Population
1940	2,50,000	1950	4,80,500
1960	5,50,300	1970	6,38,600
1980	6,95,200		

It is decided to provide water supply of 200 lpcd in the year AD 2000. By projecting the population by the incremental increase method, the total water requirement in the year 2000 will be

STEP 1

(5)

Year	population	(x) Per decade increase	(y) Incremental increase in Populatn
1940	2,50000		
1950	4,80,500	230500	
1960	5,50,300	69800	−160700
1970	6,38,600	88300	18500
1980	6,95,200	56600	−31700
		$\bar{x} = 111300$	$\bar{y} = -5796.6$

P = population in yr 2000 & per capita demand.

$\therefore$ n = $\frac{2000-1980}{10} = 2$, $P_0 = 6,95,200$

$\therefore$ $P_{2000} = P_2 = P_0 + 2\bar{x} + \frac{2(2+1)}{2}(\Sigma \bar{y})$

$= 695200 + 2 \times 111300 + 3 \times (-5796.67)$

$\boxed{P_{2000} = 743900}$

STEP 2

$\therefore Q = P_{2000} \times$ per capita demand

$= 743900 *$ 200 liter/day $/ 1 \times 10^6$

persons * litr/person/day

$Q = 148.78 \frac{ML}{day}$

$=$

* (1 Million Lt) $1 ML = 10^6 Lt$

$1 Lt = \frac{1}{10^6} ML$

Decreasing rate of growth method

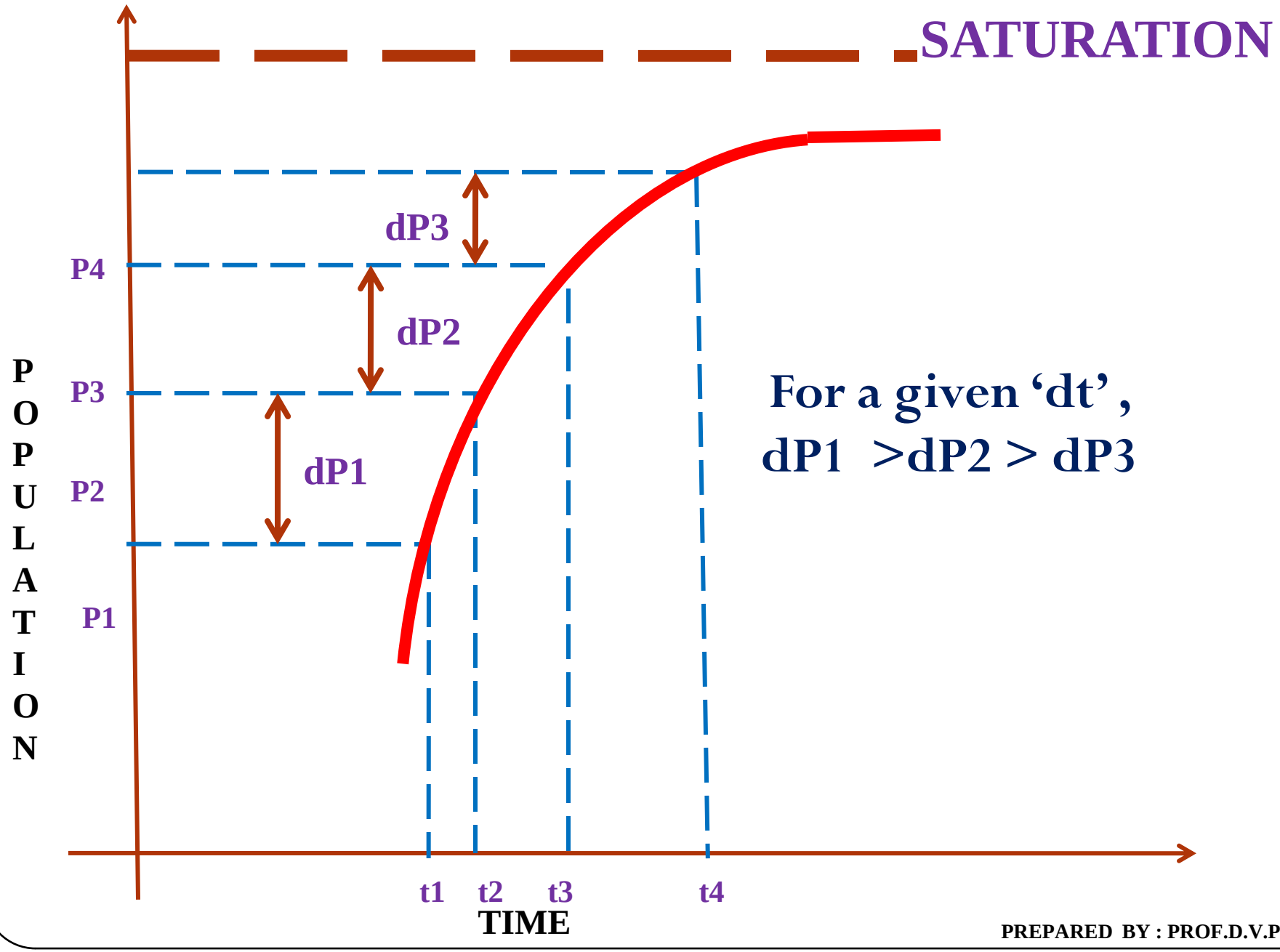
- It is rational method for the cities whose rate of increases goes on reduce , as they reach saturation.
- This method is extension of geometric increases method.
- As cities approaching towards saturation rate of increase , decreases with respect to time.

$$Let\ \bar{D} = avg.decrease\ in\ \%\ increases$$

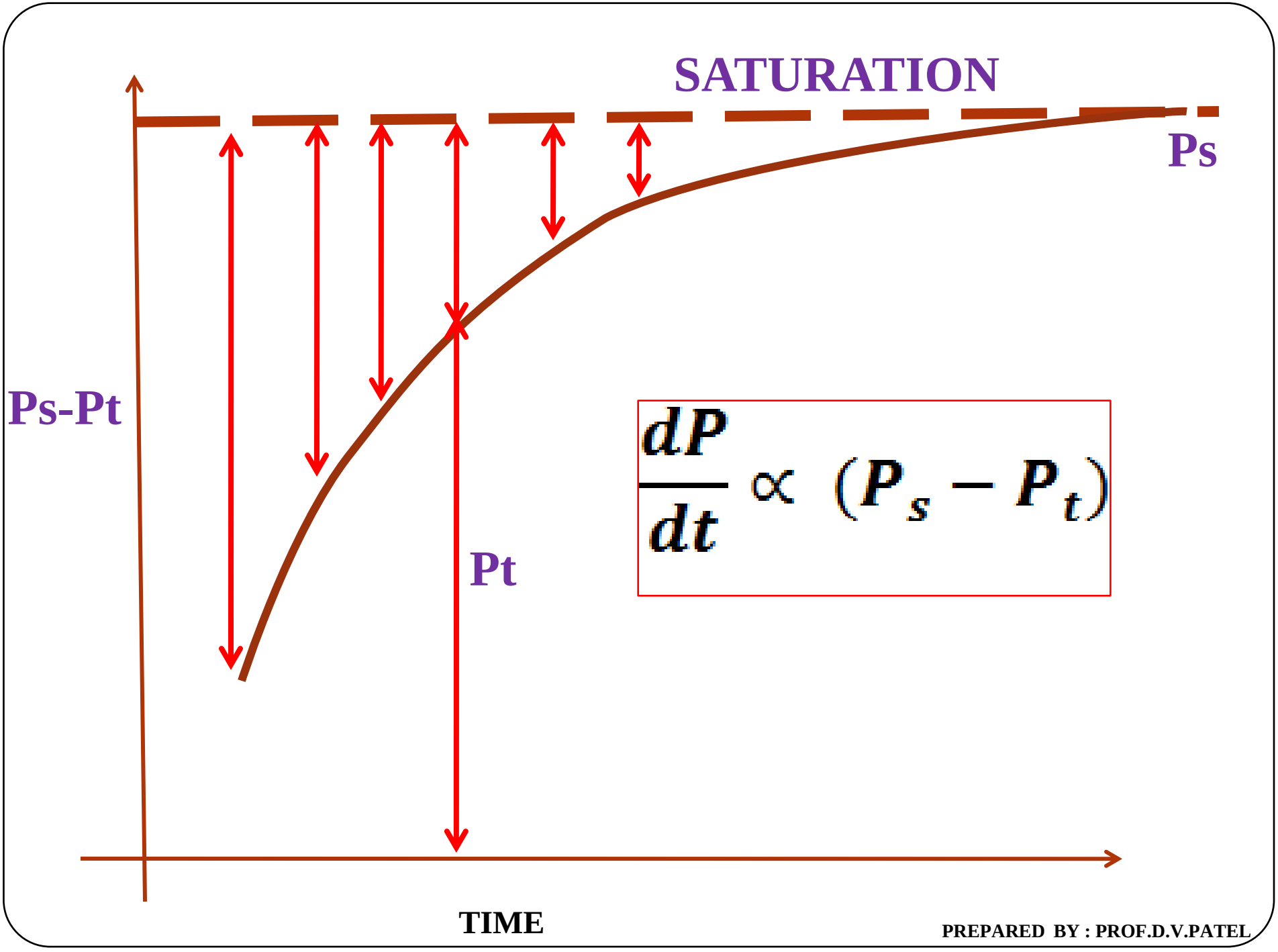
$$and\ r_0 = Latest\ known\ last\ decade\ \%\ increase$$

Population after 'n' Decade is

$$P_n = P_{n-1} * \left[1 + \frac{r_0 - n\bar{D}}{100}\right]$$



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SATURATION
Ps
Ps-Pt
Pt
$\frac{dP}{dt} \propto (P_s - P_t)$
TIME

The census record of a particular town shows the population figures as follows. Estimate the population for the year 2020 by decreasing rate growth method

Year	Population
1980	55500
1990	63700
2000	71300
2010	79500

Sol:

Year	Population	Per decade % increases in population	Decrease in % increase
1980	55500	15.31	
1990	63700	11.93	+3.88
2000	71300	11.5	+0.43
2010	79500		

$$P_0 = 79500,\quad r_0 = 11.5,\quad n = \frac{2020-2010}{10} = 1\ decade$$

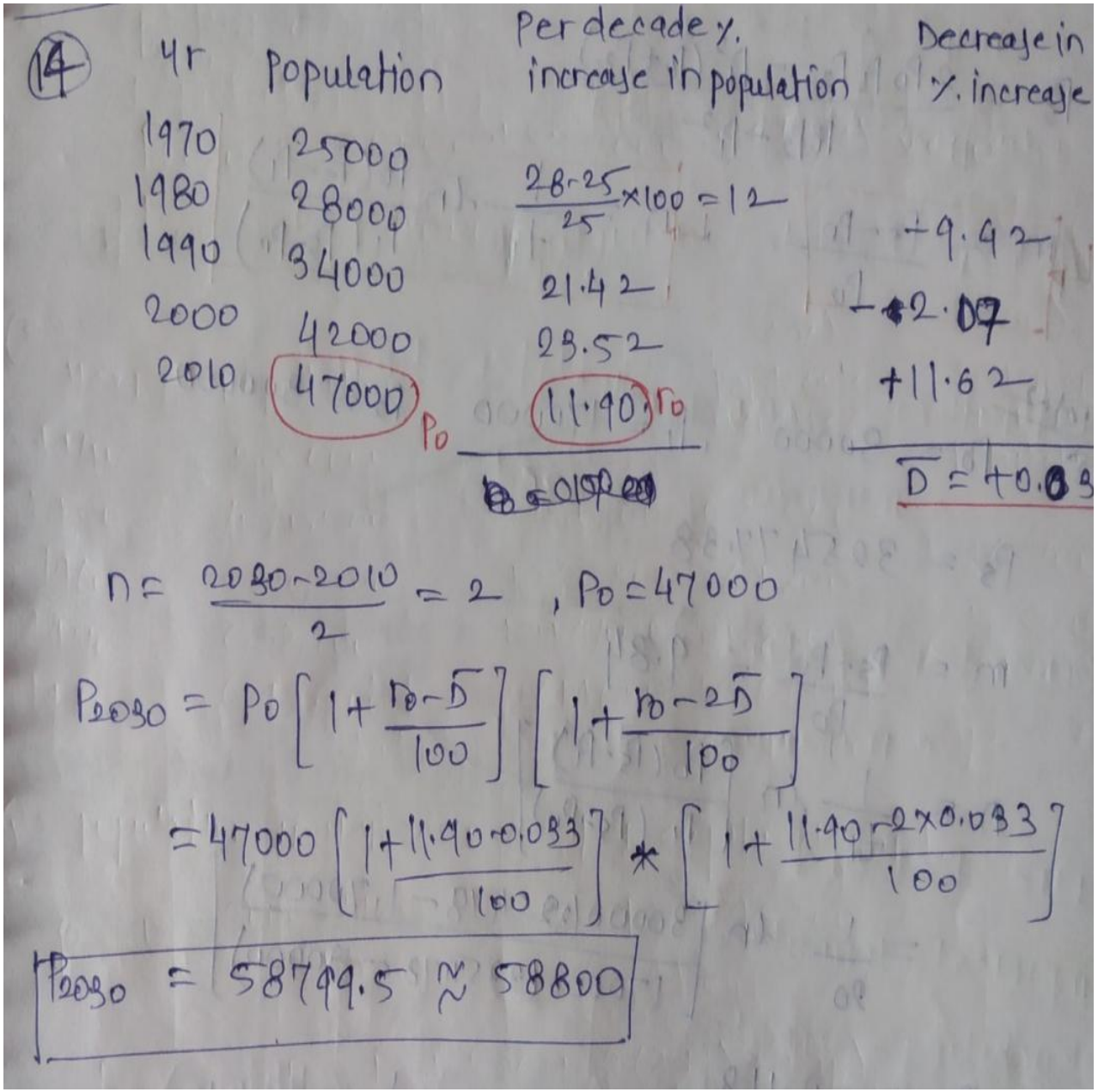
$$\overline{D} = \frac{+3.88+0.43}{2} = +2.155\%$$

$$P_n = P_0\left[1+\frac{r_0-\overline{D}}{100}\right]\ldots\left[1+\frac{r_0-n\overline{D}}{100}\right]$$

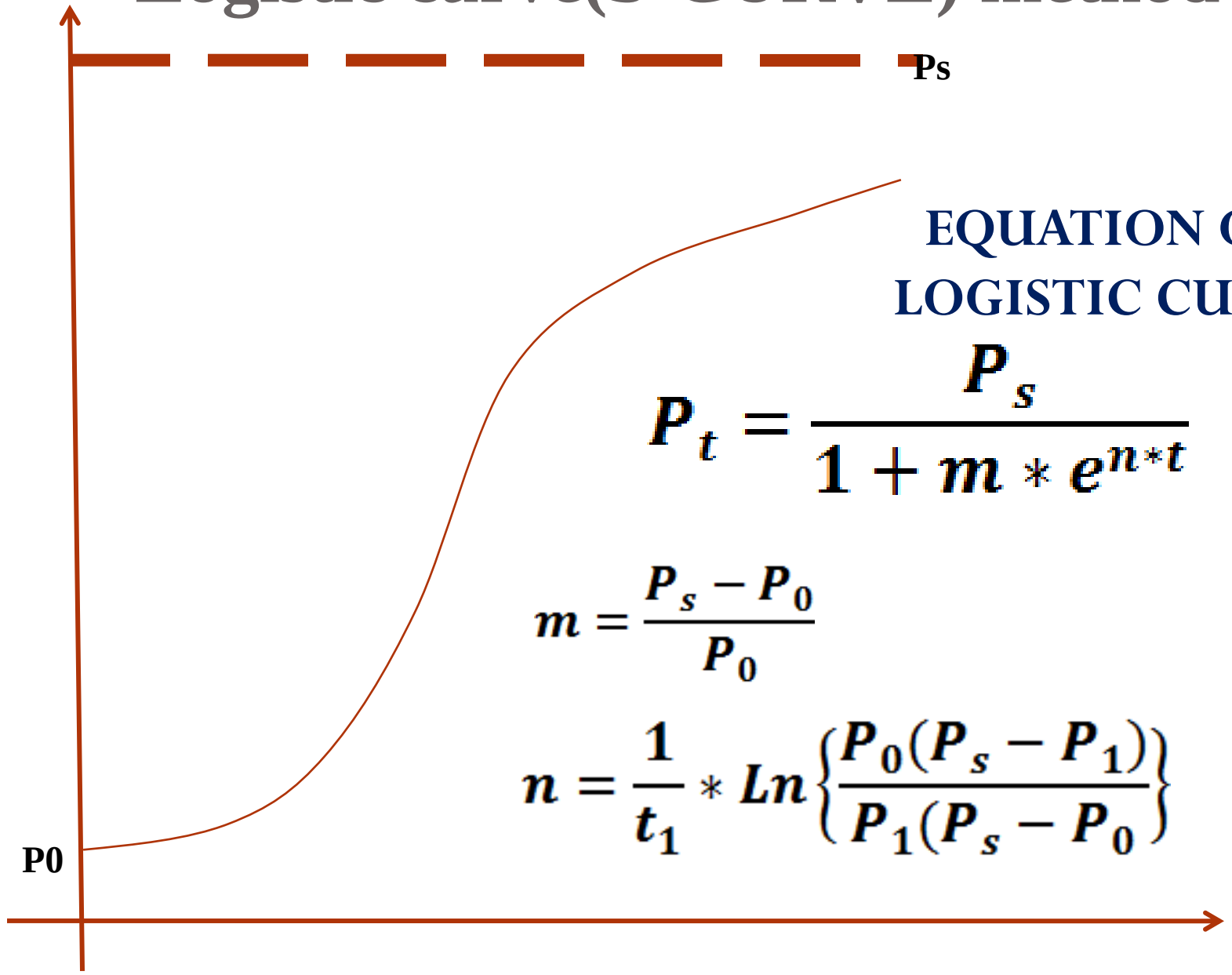
$$P_1 = 79500\left[1+\frac{11.5-2.155}{100}\right] = 86930$$

Population statistics pertaining to town given below . Find the population in 2030 by decreasing growth rate method.

Year	Population
1970	25000
1980	28000
1990	34000
2000	42000
2010	47000



Logistic curve(S-CURVE) method



EQUATION OF LOGISTIC CURVE

$$P_t = \frac{P_s}{1 + m * e^{n*t}}$$

$$m = \frac{P_s - P_0}{P_0}$$

$$n = \frac{1}{t_1} * Ln\left\{\frac{P_0(P_s - P_1)}{P_1(P_s - P_0}\right\}$$

In two period each of 20 years a city has grown from 30000 to 170000 and then 300000. Determined expected population after next 20 years.

$$P_s = \frac{2P_0P_1P_2 - P_1^2(P_0+P_2)}{P_0P_2 - P_1^2}$$

$$m = \frac{P_s - P_0}{P_0} \qquad n = \frac{1}{t_1}\ln\frac{P_0(P_s-P_1)}{P_1(P_s-P_0)}$$

Ex-2

$P_0 = 30000$, $P_1 = 170000$, $P_2 = 300000$

$P_s = 325477.98$

$m = \frac{P_s - P_0}{P_0} = 9.84$

$n = \frac{1}{t_1}\ln\left[\frac{P_0(P_s-P_1)}{P_1(P_s-P_0)}\right]$

$= \frac{1}{20}\ln\left[\frac{30000(325478 - 170000)}{170000(325478 - 30000)}\right]$

$n = -0.118$

$t_0 = 0$
$t_1 = 20$
$t_2 = 40$
$t_3 = 60$

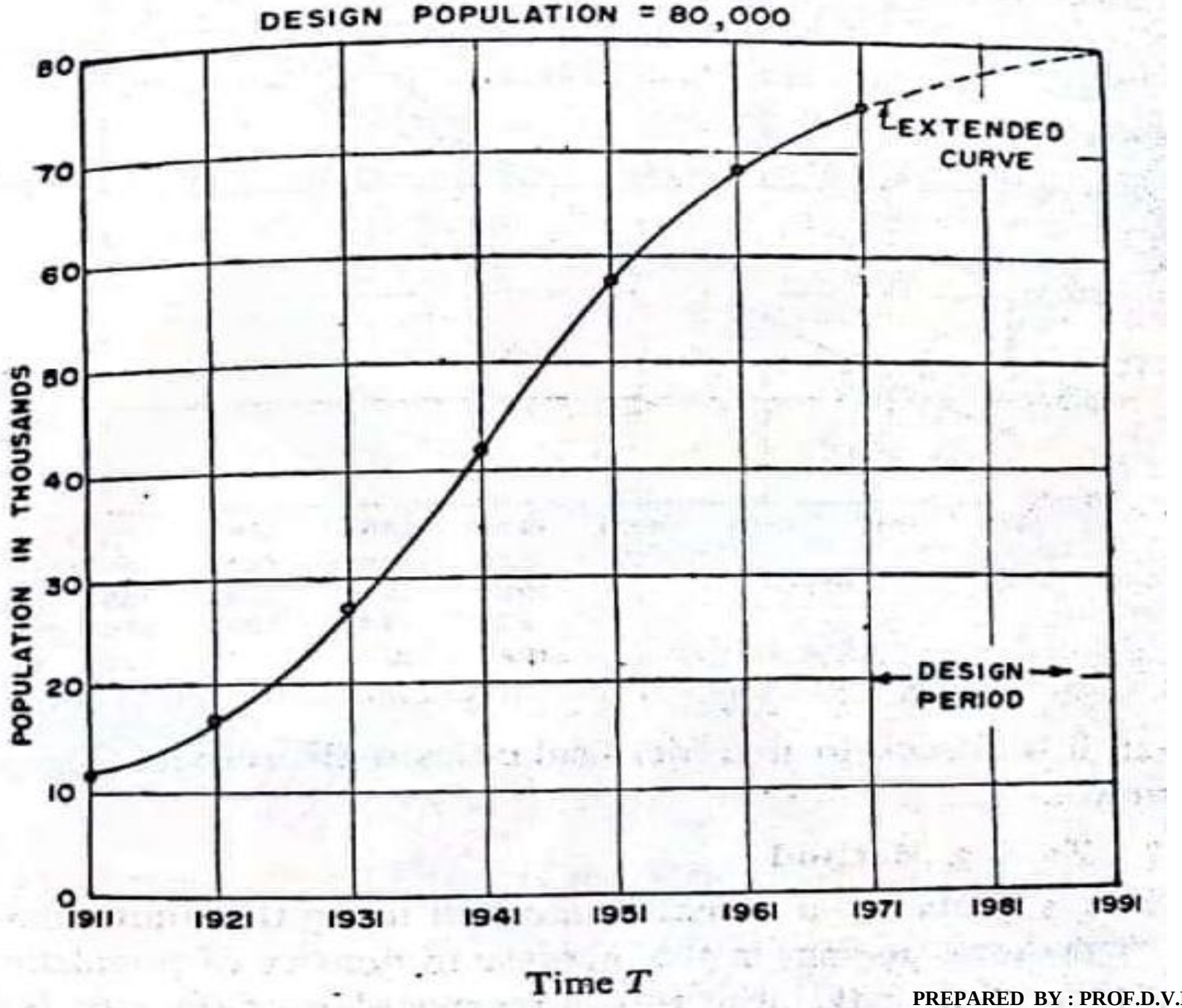
$t = 60\,yr$

$$P_t = \frac{P_s}{1+me^{nt}}$$

$P_t = 322804.19$

GRAPHICAL EXTENSION METHOD

- In this method, the populations of last few decades are correctly plotted to a suitable scale on graph.
- The population curve is smoothly extended for getting future population.
- This extension should be done carefully and it requires proper experience and judgment.
- The best way of applying this method is to extend the curve by comparing with population curve of some other similar cities having the similar growth condition.
- This gives very approximate results .
- It is very unsafe to use this method alone.


DESIGN POPULATION = 80,000
80
70
60
50
40
30
20
10
0
POPULATION IN THOUSANDS
1911
1921
1931
1941
1951
1961
1971
1981
1991
EXTENDED CURVE
DESIGN PERIOD
Time T

COMPARATIVE GRAPHICAL METHOD

- In this method, assumption is taken as that the city under consideration may develop same as the selected similar cities developed in the past.
- It is based on logical background , precise and reliable results can be obtained by this method.
- In practice however , it is difficult to find identical cities respect to population growth.

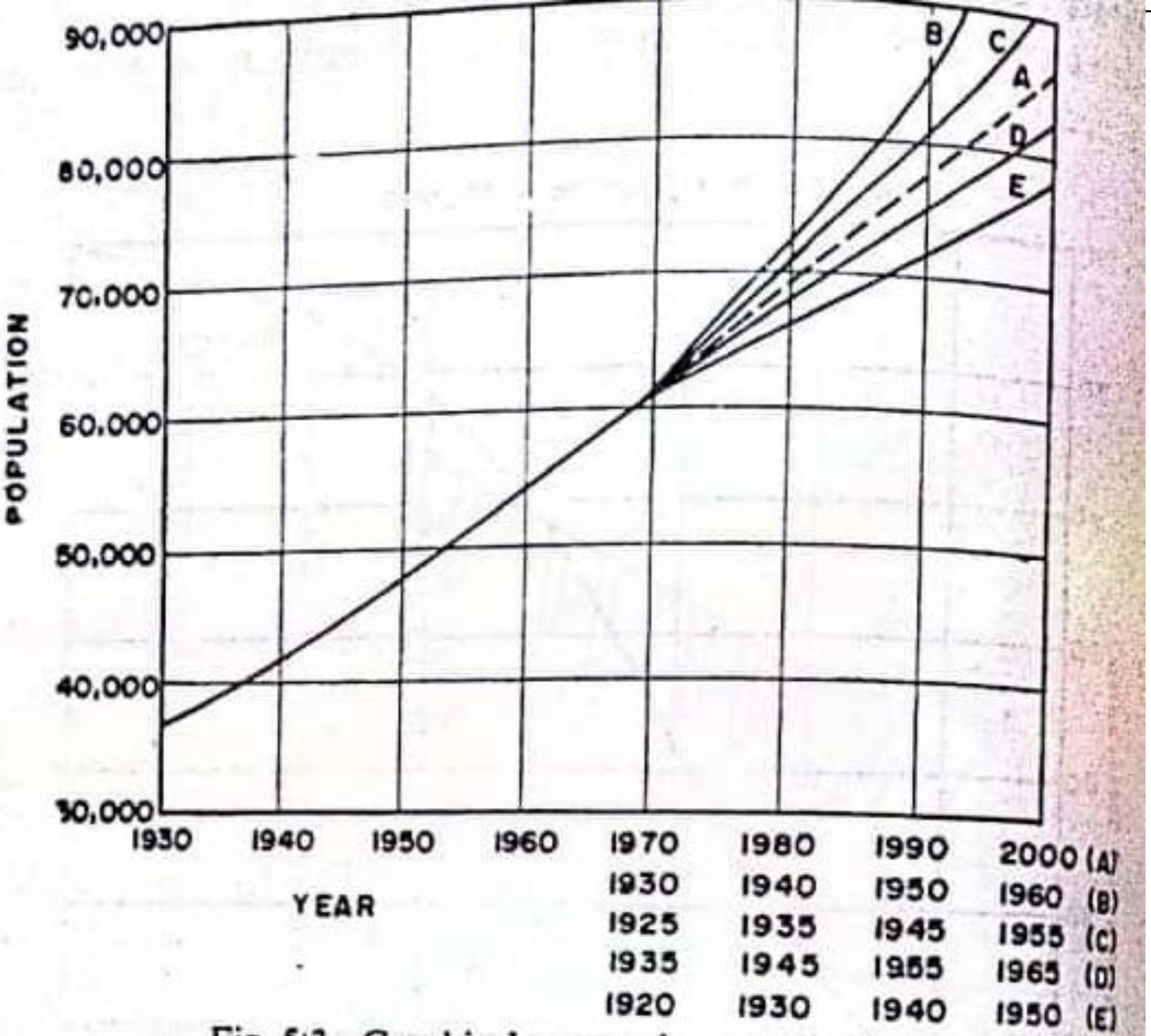


Fig. 5·3 Graphical comparison method

ZONIG OR MASTER PLAN METHOD

- This is probably a scientific method using the limitations imposed by the town planner in the increase in density of population of various parts of city.
- For this master plan is prepared for city ,dividing into different zones.
- Each zone is allowed to as per master plan only. So the future population of each zone , when fully developed can be easily found.
- This Method is more advantageous because of the fact that total water requirement of the city depends not only for domestic purposes but also for commercial ,industrial and other purposes.

FOR EXAMPLE

Sector A of residential zone has 1000 plots.

Allowing 5 persons per plot , the population of this sector when fully developed will be 1000 x 5 = 5000 persons

Table 5.1: Developed area average densities

Settlement Type	Persons per Hectare(pph) in	
	Plain Areas	Hill Areas
Small Towns	75-125	45-75
Medium Town	100-150	60-90
Large Cities	100-150	60-90
Metropolitan Cities	125-175	100-150
Megapolis	More than 200	--

Table 5.2: Land use structure for urban centres

Land use Category	Percentage of Developed Area			
	Small	Medium	Large Cities	Metropolitan Cities & Megapolis
Residential	45-50	40-45	35-40	35-40
Commercial	2-3	3-4	4-5	4-5
Industrial	8-10	8-10	10-12	12-14
Pub. & Semi Public	6-8	10-12	12-14	14-16
Recreational	12-14	18-20	18-20	20-25
Transport & Communication	10-12	12-14	12-14	15-18
Agriculture & Water Bodies	Balance	Balance	Balance	Balance
Total Developed Area	**100**	**100**	**100**	**100**

Source: UDPFI Guidelines

RATIO AND CORRELATION METHOD

- The increase in population of big cities bear a direct relationship to the population of the whole state or country also.
- In this method , local to national (or state) population ratio is determined in the previous two to four decades.
- Depending upon the conditions even changing ratio may be adopted.
- This method takes into account the regional and national factor affecting population growth .
- This method is very useful for only those area whose population growth in the past is fairly consistent with that of state or nation.

GRWOTH COMPOSITON ANALYSIS METHOD

- The change in population of city is due to three reasons birth , death and migration from villages or other towns.
- The difference between the birth rate and death rate gives the natural increase in population.
- Thus Pn = P + Natural Increase + Migration
- Natural Increase = T (Ib*P – Id*P)

Where,

T = Design (forecast) period

P = Present population

Ib= Average birth rate per year

Id= Average death rate per year